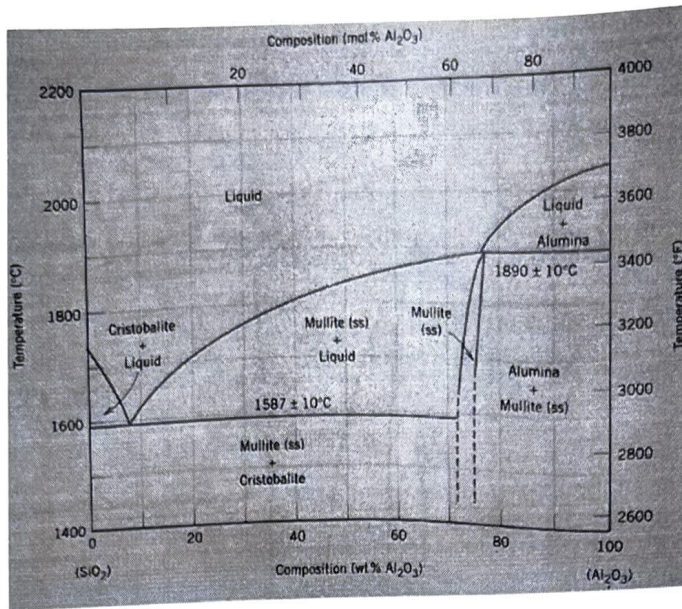


1. From the ionic radii given, estimate the coordination numbers of the cation for the following oxides: (a) MgO, (b) Al₂O₃, (c) Li₂O. $r(\text{O}^{2-})=140 \text{ pm}$, $r(\text{Mg}^{2+})=140 \text{ pm}$, $r(\text{Al}^{3+})=54 \text{ pm}$, $r(\text{Li}^{+})=76 \text{ pm}$. (9 points)
2. Lithium oxide forms in the antifluorite structure (Attached ionic crystal Radii Table for your reference).
 - (a) Compute the value of the lattice constant.
 - (b) Calculate the density of Li₂O.
3. Classify the following elements as modifiers, intermediates, or network formers (glass formers) in connection with their use in forming oxide type glasses: Si, Na, B, Ca, Al, P, K, Ba. (12 points)
4. We produce good chemical resistance in a glass when we introduce B₂O₃ into silica. To assure that we have good glass-forming tendencies, we wish the O:Si ratio to be no more than 2.5, but we also want the glassware to have a low-melting temperature to make the glass-forming process easier and more economical. Design such a glass. (12 points)
5. Calculate the number of Schottky defects per cubic meter in potassium chloride at 500°C. The energy required to form each Schottky defect is 2.6 eV, while the density for KCl (at 500°C) is 1.955 g/cm³. (10 points)
6. From the phase diagram of Al₂O₃–SiO₂, (15 points)
 - a. for the following pair of compositions, which would you judge to be more desirable refractory ? Justify your choice.

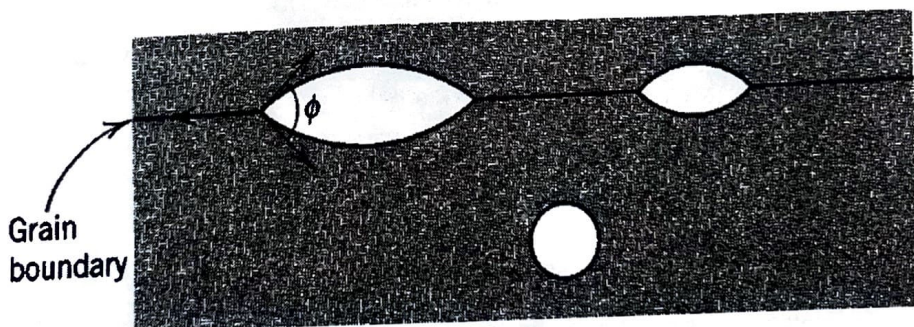
20 wt% Al₂O₃–80 wt% SiO₂

25 wt% Al₂O₃–75 wt% SiO₂
 - b. Compute the mass fractions of liquid in the 45 wt% Al₂O₃–55 wt% SiO₂ fireclay refractory materials at 1600 °C (2910 °F).



7. During sintering of polycrystalline materials, we often end up with pores of an assortment of sizes, located both within the grains and at grain junctions, as shown schematically here:

- (a) Which pore will disappear first, and why? Which pore disappears last, and why? (7 points)



- (b) If the three pores shown above are close enough for atom diffusion to take place readily, will there be any mass transported between the pores? (Aside from the tendency of each to shrink.) Show the direction of mass transport, and explain (6 points).
8. What is the driving force for sintering? List three types of sintering techniques for ceramics and how to suppress grain growth during sintering? (9 points)

The edge length of the unit cell of LiCl (NaCl-like structure, FCC) is 0.514 nm or 5.14 Å. Assuming that the lithium ion is small enough so that the chloride ions are in contact, as in